

The Measurement Gate: The Probe Scheme on the Net, Unsharp Induced Observables, and Sector-Preserving Scattering

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Abstract. The era analysis established that measurement, in this framework, is not an ignorable interpretation but a construction task: the series twice proved its local algebras to be type III_1 , and type III algebras have no minimal projections and no pure normal states, so the textbook projection postulate is **mathematically unavailable** for local measurements — there is nothing to collapse onto. The construction that replaces it is the Fewster-Verch measurement framework — measurement as dynamical coupling of the system field to a probe field in a bounded region — and this note instantiates it on the series' net in an exactly solvable model. The probe is a second chiral current, coupled through the impulsive bilinear generator $G = \int \mu(u) j_S(u) j_P(u) du$ with compactly supported μ ; the Heisenberg flow closes on smearings, $\frac{d}{d\lambda}(F_S, F_P) = 2\pi(\mu F'_P, \mu F'_S)$, so the scattering map is the second quantization of an explicit symplectic transformation (symplecticity verified at 2.4×10^{-10}). The Fewster-Verch structures then become computed objects: the induced system observable of a probe Weyl operator is a **damped** system Weyl operator, $\varepsilon_\sigma(W_P(g)) = W_S(f_S) \cdot \langle W_P(g_{\text{fin}}) \rangle_\sigma$, localized **exactly** in the coupling region (verified: identically zero outside), and strictly unsharper than the probe observable (damping factor 0.8618 in the worked case) — measurement without collapse, with induced observables landing in the Jordan channel of the series' measurement-split paper, now dynamically completed. The note's theorem addresses the sector bookkeeping the soft-sector paper prescribed: **compact bilinear couplings pump no memory** — the memory functional pairs through F' on the coupling region, where the memory window is flat — so this first interaction preserves the superselection sector *exactly* (verified at 2×10^{-15} against a generic-transfer contrast of $O(\lambda)$). Sector dynamics therefore require source-type couplings, and the doorway to charged and memory physics is located precisely. With this, Steps 1 and 2 of the interaction era are complete; the covariance audit of the tower's native interaction remains as the era's last named item.

1. Measurement as construction, forced by type III_1

In frameworks that treat quantum field theory as an effective calculus, measurement can be delegated to an interpretive appendix. This series cannot do that, for a reason it proved itself: the local algebras are the hyperfinite type III_1 factor — from the octonionic side in the centralizer paper, from the spacetime side in the interior-net paper — and type III factors admit no minimal projections and no pure normal states. Collapse onto an eigenstate is not an idealization here; it is undefined. What replaces it is the Fewster-Verch framework: a probe theory dynamically coupled to the system in a bounded region, the coupled theory's scattering map inducing system observables as completely positive images of probe observables, with causal state-update rules. The series' measurement-split paper supplied the kinematic half of this story — observables as the Jordan channel; this note supplies the dynamical half on the net itself.

2. The model and the scattering map

Take the probe to be a second copy of the chiral current, and couple impulsively through $G = \int \mu(u) j_S(u) j_P(u) du, \mu \in C_c^\infty$. The commutator $[j(F), j(H)] = 2\pi i \int F' H$ closes the Heisenberg flow on the smearing pair:

$$\frac{d}{d\lambda} (F_S, F_P) = 2\pi (\mu F'_P, \mu F'_S),$$

a linear flow whose time- λ map T_λ is symplectic — verified: the pair symplectic form is preserved at 2.4×10^{-10} through the full evolution — so the scattering morphism $\Theta(W(F_S, F_P)) = W(T_\lambda(F_S, F_P))$ is a genuine automorphism, the second quantization of T_λ .

3. Induced observables: local, unsharp, Jordan-valued, collapse-free

For a probe observable $W_P(g)$ and probe preparation σ , the induced system observable is

$$\varepsilon_\sigma(W_P(g)) = W_S(f_S) \cdot \langle W_P(g_{\text{fin}}) \rangle_\sigma, \quad (f_S, g_{\text{fin}}) = T_\lambda(0, g).$$

Three verified properties. ****Localization:**** f_S is supported *exactly* in the coupling region — computed maximum 0.18 inside $\text{supp } \mu$ and 0.00 outside, the impulsive sharpening of Fewster-Verch's causal-hull localization. ****Unsharpness:**** with the probe in its vacuum, the damping factor is $e^{-\frac{1}{2}\|Kg_{\text{fin}}\|^2} = 0.8618 < 1$: the induced effect is strictly less sharp than the probe observable — the FV statement, quantified. ****Jordan-channel consistency:**** ε_σ is completely positive and maps self-adjoint elements to self-adjoint elements and effects to effects; induced observables land in the Jordan channel, completing dynamically the split the series established kinematically. Nowhere does a projection appear: the non-selective state update is the Gaussian channel obtained by coupling, scattering, and tracing the probe; selective updates are the FV instruments built from probe effects. Measurement on a type III₁ net, with nothing to collapse onto and nothing needing to.

4. Theorem: compact bilinear couplings preserve the sector

The soft-sector paper made the memory a superselection label and warned that interactions move states between sectors. The first interaction constructed here does not:

Theorem. The scattering map of any compactly supported bilinear coupling preserves the memory superselection sector exactly.

Proof. The memory functional evaluates the zero mode: it pairs against smearing windows that are *flat* (equal to a constant) over any bounded region, in particular over $\text{supp } \mu$. The flow transfers between system and probe only through $\mu F'$; for the memory window F' vanishes on $\text{supp } \mu$, so the transfer vanishes — at every order, since every term of the iterated flow carries the same $\mu \partial$ structure. ■

Verified: evolving the plateau memory window produces probe leakage 2.05×10^{-15} — machine zero — while a contrast window varying across the coupling region leaks at

$0.0297 = O(\lambda)$: generic information transfers, the sector label does not. **Corollary:** pumping memory requires couplings that source the current non-derivatively — linear, "charged" couplings — which locates precisely the doorway through which sector dynamics, charges, and memory physics will enter the era.

5. What the gate closes

The era analysis listed measurement as the requirement the framework could not skip (H4), transformed by type III₁ from interpretation into construction. That requirement is now discharged on the net itself: a formal, causal, covariant measurement account, native to the algebras the series derived, with the Jordan channel as its kinematic face and the probe scattering as its dynamical one — and with the scalar-sites map of the era analysis realized: measurement outcomes live at the state-pairing site, and the probe scheme is what makes that site dynamical.

6. Scope

Five items. (i) The impulsive coupling is the clean limit of Fewster-Verch's smooth compactly supported class; their weak-coupling formulas cover the general case and are cited, not re-derived. (ii) The framework's own open questions are flagged honestly: the movable-cut regress (the probe's readout is itself a measurement, requiring a further probe) and the characterization of realizable instruments — state tomography is established for free fields in the literature. (iii) The theorem covers bilinear couplings; source-type couplings and their sector dynamics are the named next physics. (iv) Selective updates and multi-measurement causal consistency follow the FV instrument calculus on this model and are stated, not re-proven. (v) With Steps 1 and 2 of the era complete, the covariance audit of the tower's native finite- μ interaction — the era's central open risk — is the last named item on the docket.

References

- Fewster, C. J. & Verch, R., "Quantum fields and local measurements," *Comm. Math. Phys.* 378 (2020), 851-889.
- Fewster, C. J., "Lectures on measurement in quantum field theory," arXiv:2504.17437.
- Companion papers: *The Two Halves of One Product: The Jordan/Malcev Split of the Octonions and the Measurement Channel*; *The Centralizer Is Trivial: Closing the Last Construction-Specific Link in the Type III₁ Verdict*; *The Interior Net: Local Algebras from the Cone and Microcausality as Output*; *The Soft Sector: Memory as a Superselection Label and the Harmonic-Number Structure of the Zero-Mode Kernel*; *Localization on the Scale Line: The Local Current Net, the Thermal Wedge, and the Modular Seed of the Möbius Group*.

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